CST2550 - Coursework 2

Project Requirements

Group Members		
Student ID	**Student Name/Surname**	**Role**
M01015301	Nunkoo Namrata-Riya	Scrum Master
M01038115	Hau Yam Seung Mélanie	Secretary
M01014266	Coret Mégane	Developer 1
M01026956	Ramtohul Syrielle	Developer 2
M01015305	Sedoyeka Rayan	Tester

Submission Date: 15/04/26
Lab Tutor: Dr. Gina Soopramanien

Table of Contents

1. Introduction *3*

2. Project Requirements *3*

2.1 Functional Requirements **3**

2.2 Non-Functional Requirements **5**

3. System Structure and Pages *5*

4. Use Case Diagram *6*

5. References *7*

Table of Figures

Figure 1: Use Case Diagram **6**

1. Introduction

This document outlines the functional and non-functional requirements of the Football Betting System, as well as the use case showing the interaction of the users with the system.

2. Project Requirements

Analysing the project requirements is essential to understand what the system should do and how it should perform to ensure that it behaves as expected. For a structured analysis, requirements are broken down into functional and non-functional requirements.

Functional requirements define what the Football Betting System should do by defining the features and behaviour of the system during interaction with users. Non-functional requirements, however, define how the system is expected to behave, focusing on the security, performance, usability, and reliability. (GeeksForGeeks, 2026)

2.1 Functional Requirements

The system shall:

- Allow users to login using email and password.
- Allow users to register a new account with their first name, last name, date of birth, email, and password.
- Prevent users from logging in if they are banned or suspended.
- Display error messages for invalid login or registration attempts.
- Allow users to logout.
- Allow users to view or edit their profile details.
- Prevent users from modifying their date of birth.
- Allow users to edit their password.
- Validate password during registration and password change.
- Enforce password validation requiring a minimum of 8 characters with at least one number, one uppercase letter, and one special character.
- Display error messages indicating which password requirements have not been met.
- Display upcoming football matches.
- Allow users and admin to search matches by team name.
- Allow users and admin to filter matches by league.
- Prevent users from betting on matches that have already started or have been completed.
- Allow users to place bets on a match.
- Update existing bet if same match and bet type already exists in slip
- Retrieve and display the odds for each bet.
- Allow users to view their current unpaid bet slip.
- Allow users to remove bets from their unpaid slip.
- Allow users to enter the stake
- Allow users to confirm and pay for a bet slip.

- Prevent bet payment if balance is insufficient
- Deduct amount from user wallet upon payment
- Calculate and display potential payout for user's bet slip.
- Calculate and display total odds for user's bet slip.
- Display bet outcomes to users after the match is completed.
- Allow users to view their bet history.
- Allow users to filter their bet history by date.
- Allow users to filter their bet history by status (won, lost or pending).
- Allow users to view results of each bet in their history slips.
- Allow users to claim bet slip they have won.
- Update user's wallet balance after payout.
- Display users' current wallet balance.
- Allow users to deposit money in their wallet.
- Allow users to withdraw money from their wallet.
- Validate deposit and withdrawal amounts
- Prevent users from withdrawing more than their wallet balance
- Allow admin to view all users.
- Allow admin to suspend, block or unsuspend users.
- Prevent admin from modifying already suspended users
- Allow admin to view all matches
- Allow admin to view results of completed matches.
- Allow admin to add new matches.
- Generate match results when admin adds a match
- Generate odds when admin adds a new match
- Prevent admin from adding a match which starts in less than 5 minutes
- Allow admin to view total revenue
- Allow admin to view total number of active users
- Allow admin to view total number of bets placed
- Allow admin to view deposits total
- Allow admin to view withdrawals total
- Allow admin to view a line chart for profit and loss over the last 12 months
- Allow admin to view a grouped bar chart for transaction volume over the last 12 months
- Allow admin to view a donut chart for the bet outcome
- Allow admin to view a generated financial report
- Automatically update match status
- Automatically evaluate bets after match completion and update their results
- Automatically update bet slip status based on the outcomes of associated bets
- Generate odds for bets that do not have predefined odds when a user places a bet
- Generate odds for all matches without odds on system startup

2.2 Non-Functional Requirements

The system shall:

- Encrypt all user passwords using hashing algorithms for password protection.
- Ensure the system is user friendly, with a simple interface that is easy to navigate.
- Display clear error messages when an action fails, or input is invalid.
- Display confirmation messages after every successful action.
- Ensure the system remains stable and reliable.
- Restrict admin functionality through authentication.
- Ensure data consistency and integrity during transactions

3. System Structure and Pages

The system consists of multiple pages and popups to support betters and the administrator. All users access the system through the landing page and access to specific features is determined based on user roles after authentication.

Common Pages:

- **Landing Page**: Serves as entry point to the system, providing access to sign up or log in.
- **Sign Up/Log In Page**: Allows users to create an account or authenticate into the system

User Pages

- **Main Page**: Displays upcoming matches and enables users to browse matches and place bets
- **Profile Page**: Display user information, wallet balance and provides navigation to the bet history page.
- **Bet Slip Page**: Display selected unpaid bets, allowing users to remove bets, enter stake amount and confirm bet slip. Once confirmed, the bet slip cleared and saved to database.
- **History Page**: Display users' previous bet slips, their status and the outcome of each associated bet.

Admin Pages

- **Users Page:** Display all users and allow admin to modify user status.
- **Matches Page:** Display all matches and display match results for completed matches.
- **Add Match Page:** Allow admin to add new scheduled matches.
- **Financial Page:** Display financial summaries with graphs and enable admin to generate a financial report using an AI Agent.

Popups:

- **Bet Details Popup**: Displays information about bets from a selected slip from history page.
- **Money Withdrawal and Deposit Popup:** Allows users to deposit or withdraw money.
- **Profile Edit Popup:** Enables users to update their profile details.
- **Password Change Popup:** Allows users to update their account password.
- **Logout Popup:** Prompt users to confirm when logging out or exiting application.

4. Use Case Diagram

The use case diagram in Figure 1 shows how the actors, namely the administrator, AI agent and users, interact with the system by illustrating high-level functionality (GeeksForGeeks, 2026).

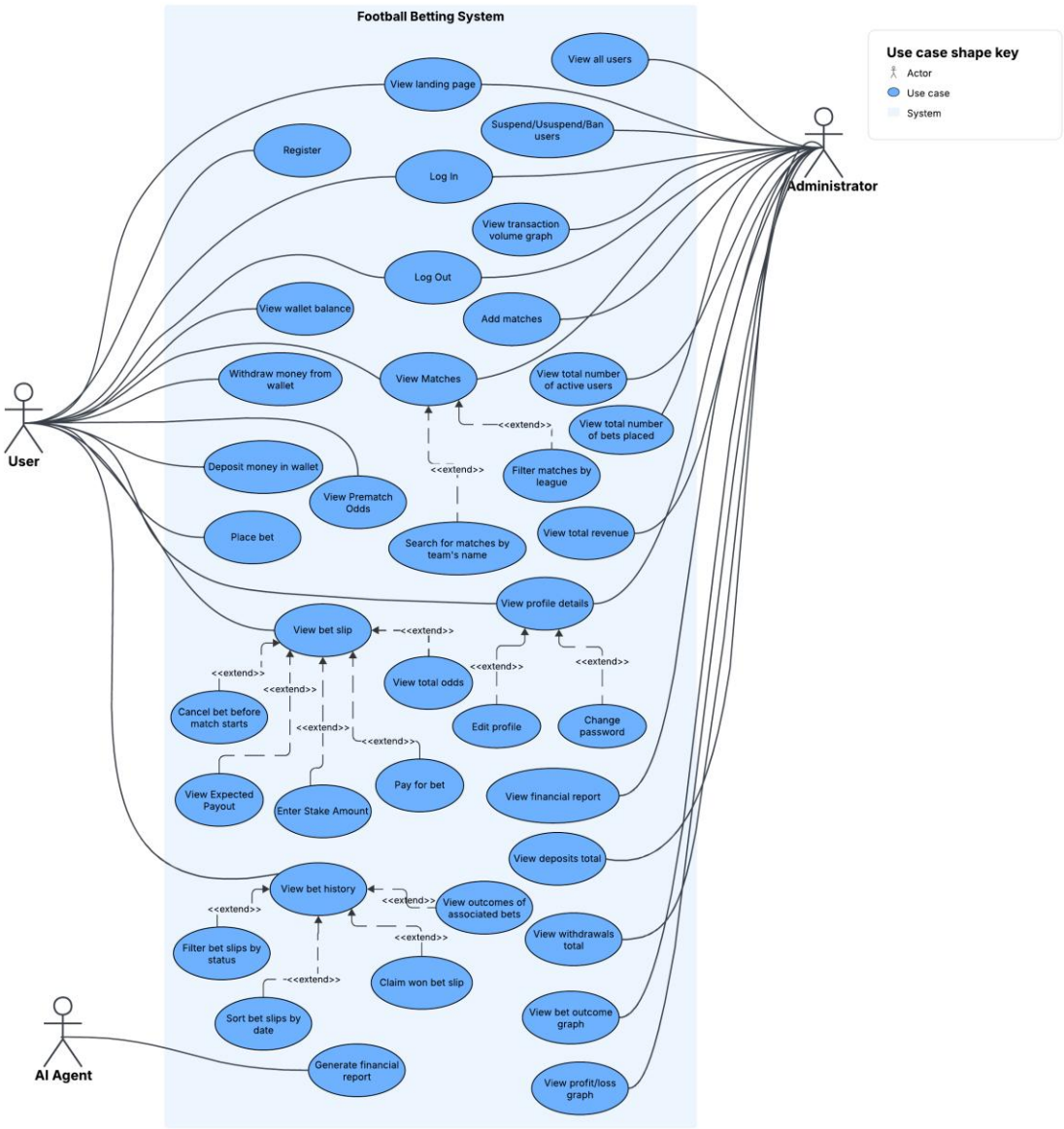

Figure 1: Use Case Diagram

5. References

GeeksForGeeks (2026) *Functional and Non Functional Requirements*. Available at: https://www.geeksforgeeks.org/software-engineering/functional-vs-non-functional-requirements/ (Accessed: 8 April 2026).

GeeksForGeeks (2026) *Use Case Diagram - Unified Modeling Language (UML)* Available at: https://www.geeksforgeeks.org/system-design/use-case-diagram/ (Accessed: 8 April 2026).